
Pev: A Calibrated Fast-Decision Model for Personal Agents

LoRA SFT of Qwen3.8-27B on a pre-registered benchmark built from real public data

EnvLoop Research
research@envloop.ai

Abstract

Personalized agents (assistants that keep a long-term memory of one user and act on that user’s behalf across shopping, travel, email and calendar) are quickly becoming a product category. Their bottleneck is less open-ended generation than *personal judgement*: deciding, many times per request, what this particular user would want. Asked to “re-order the dog food”, such an agent has to infer that the user switched brands after an allergy, that the new \$54 bag now exceeds the user’s \$50 approval rule, that an address the user asked it to forget must not be reused, and whether the request is worth interrupting the user for. Each decision is small, but errors cost in both directions: an agent that asks too often is useless, and one that acts wrongly is unsafe. We cast these judgements as seven typed question families and train *Pev-27B*, a fast decision model that answers each with a calibrated probability from a single forward pass: a rank-16 LoRA on Qwen/Qwen3.8-27B, supervised on a single option-label token and read out as a temperature-calibrated distribution over the options. We also release *Pev-Bench*, generated from real public behaviour (Amazon Reviews 2023, UCSD Google Local) and real distractor text (Enron, OpenFlights), with labels fixed by a knowledge base before any text is rendered, exact label and answer-position balance, verbatim anchor checks and user-disjoint splits. Under a pre-registered protocol, the frozen model passed a one-shot HIDDEN gate: family-macro accuracy rose from 0.754 to 0.905 (+15.1 points, 95% CI [+13.5, +16.8]), safety false negatives fell, and the share of questions it can answer alone within a 5% error budget rose from 0.54 to 0.93. On a public TEST set of 720 new users, half rendered by gpt-6-astra and half by Claude Opus 5.5, it reaches 0.915, ahead of gpt-6-astra (0.873), Kev-27B (0.823), Jev (0.788) and its base model (0.762), overall and on both halves (Holm-adjusted $p = 5 \times 10^{-4}$ for every comparison). Predicting which option a user will pick from real behaviour remains unsolved for every model, and gpt-6-astra keeps slightly fewer safety false negatives on two families.

Technical report. Code: <https://github.com/EnvLoop/Pev>; data: EnvLoop/Pev-Bench and weights: EnvLoop/Pev-27B-LoRA on Hugging Face (gated).
This report, its tables and figures are licensed under CC-BY-4.0.

1 Introduction

Personalized agents. Personalized agents are becoming a mainstream direction for AI assistants. Systems in the style of Muse [12] keep a long-term memory of one user, act across that user’s services (shopping, restaurants, travel, email, calendar), follow user-defined approval rules for sensitive actions, respect a privacy level per fact and recipient, and forget on request. As these agents move from answering questions to taking actions, what matters most is no longer fluent generation but *personal judgement*: doing what this particular user would want.

Personal judgement is hard: an example. Consider a user who tells their assistant “re-order the dog food”. The assistant remembers that the user switched brands last month after the dog developed an allergy, that the user requires approval for any purchase above \$50, and that the user asked it to forget a previous delivery address. Two bags are in stock: the old brand at \$38 and the new one at \$54. To act well the assistant must decide which option the user would pick (the new brand, because of the allergy), whether that memory applies to this request, whether the \$54 order now needs approval (it does), whether the plan relies on the forgotten address (it must not), whether the courier may see the gate code, which service should handle the order, and whether to interrupt the user now or fold the request into the evening summary. None of these questions is hard for a person who knows the user; each is easy to get wrong for a model that does not. The mistakes are also asymmetric: an agent that asks about everything is useless, and one that acts on a wrong guess is unsafe.

What is missing. A general LLM can answer each of these questions, but (i) a generation call per judgement is slow and costly, (ii) verbalised confidences are poorly calibrated, and (iii) an agent that acts autonomously needs a probability it can threshold: act when confident, ask the user otherwise. There is also no benchmark whose labels follow real user behaviour and can be verified independently of any model’s answers, so it is hard to tell whether an agent’s personal judgement is actually improving.

This work. We cast personal judgement as seven typed question families (`pick_option`, `apply_memory`, `needs_approval`, `forgotten_violation`, `share_ok`, `notify_level`, `route`) and train a model that emits one distribution per question from a single forward pass, calibrated with one temperature fitted on held-out data [3]. Besides accuracy we measure *automation coverage*: the share of questions the model may answer alone while the error on those answers stays within a 5% budget. On a public TEST set of 720 new users, Pev-27B reaches a family-macro accuracy of 0.915, ahead of gpt-6-astra (0.873), Kev-27B (0.823), Jev (0.788) and its own base model (0.762), and can answer 94% of questions alone within the 5% error budget; predicting which option a user will pick from their real behaviour remains largely unsolved for every model we test.

Related systems. *Kev* [14] (jaredpalmer/kev, pinned at 0fe8fc97) defines the typed decision format we adopt (questions of type `noul` (yes/no), `choice` and `score` over a natural-language state, with labels and soft targets) and trains a rank-16 LoRA with a pointer head whose temperature is fitted on validation data. Its released checkpoint Kev-27B [15] (jaredpalmer/kev-27b@01b81998) is a reference here, and its `decision-v7` test suite is our regression guard. *Jev* [20] (jev-1.13.0) is TypeSafe’s “System One” decision model, queried through the official `typesafe-sdk` [19]; it is also a reference only. Following amendment 2 we use neither Kev’s training code nor its pointer head: the candidate is plain SFT on the base model’s own next-token distribution, so B0 and the adapter differ only by the LoRA weights. gpt-6-astra renders our data and is a third reference. The name Pev follows the naming of earlier decision models such as Kev and Jev; Pev is developed independently and is not affiliated with either project.

Contributions. (1) A benchmark generator in which labels come from a knowledge base of real user histories plus seeded rules, never from a model’s answers, with exact balance and verbatim anchor verification (§2). (2) A pre-registered, noise-aware evaluation protocol with a one-shot `HIDDEN` gate and validity checks (§3). (3) A single-label-token LoRA recipe whose general-decision data mix removes the regression on an external guard (§4). (4) Results on DEV, `HIDDEN` and a public TEST set with two renderers (§5), and an account of what does not work (§6). Code, data, weights and the pseudonymization key are released (§7).

2 Task and benchmark

2.1 Records and question families

A record is one `state` (rendered text: what the assistant remembers about the user, the current request, candidate actions and optional distractor text) with up to three typed questions, each from a different family. Labels follow Kev: the option name for `choice`, `true/false` for `noul`, the zero-based level for `score`. Table 1 lists the seven families; Appendix C defines them in full.

Table 1: Question families. Labels are computed from the knowledge base, never from a model’s answers.

Family	Type	Label source	Variants
<code>pick_option</code>	choice	Real behaviour: an item the user rated ≥ 4 <i>after</i> the memory cut-off; distractors are the user’s real later item rated ≤ 2 (when balanceable) and untouched items of the same category	override, removed (soft)
<code>needs_approval</code>	noul	Spend threshold in force at that time, new-recipient rule, irreversible-action rule	override, removed (soft 0.5)
<code>apply_memory</code>	noul	Whether the fact’s category matches the request (guards against over-personalisation)	clean
<code>forgotten_violation</code>	noul	Whether the candidate action relies on a fact the user asked to forget	removed (contrast)
<code>share_ok</code>	noul	Recipient clearance $\geq$ the fact’s privacy level	override, removed (soft 0.5)
<code>notify_level</code>	score	Urgency capped by the type maximum, quiet hours and the meeting cap, with VIP bypass	override, removed (tied soft)
<code>route</code>	choice	The needed service if connected, else <code>ask_user</code> (also for OpenFlights city names ambiguous across countries)	override

Table 2: Public sources of the knowledge base.

Source	Use	Selection
Amazon Reviews 2023 (McAuley Lab, UCSD) [5]	Product histories (memory, real later choices)	5 categories (Grocery and Gourmet Food, Pet Supplies, Baby Products, Appliances, All Beauty); 8,000 users sampled
Google Local 2021 (UCSD) [8, 22]	Restaurant histories	District of Columbia, Delaware, Rhode Island (10-core files); 38,263 eligible users
OpenFlights [13]	Airports, airlines and routes for travel routing	7,698 airports, 6,162 airlines, 67,663 routes
Enron email corpus (CMU) [7]	Buried distractor text only	20,000 emails sampled; addresses, phone numbers and URLs replaced by placeholders

Hard variants: evidence *buried* in 1k–6k tokens of distractor text (whole state, $p = 0.25$), preferences *overridden* by later updates, and evidence *removed* (the question then carries a soft label). A question whose soft label has no unique argmax is *ambiguous*: it enters Brier/ECE/NLL only, never accuracy, McNemar, safety, automation or the SFT rows.

2.2 Knowledge base from real public data

The normalised tables (Table 2) hold 90,922 items and 881,370 interactions. The KB population is 3,000 users (1,500 product, 1,500 restaurant; 8–30 distinct items, $\geq 80\%$ of them with review text, and at least one later rating ≥ 4 and one ≤ 2), ordered by a seeded hash. Each history is sorted by time; the first 60% becomes memory and the rest are real later choices. Memory facts (25,748) are extracted from early reviews, 90.6% of them by gpt-6-astra, which must return a verbatim evidence span or the rule extractor is used instead. Approval rules, privacy levels, contacts, calendars, notification rules and forget requests are **synthetic**, seeded by user id; they sit next to real review histories and are not attributes of the real reviewers.

2.3 Question generation, balance and rendering

Each family is a pure function of the KB that returns state lines, one pending item, one question, its label and its *anchors* (strings that must, or must not, appear in the final text). Up to three compatible fragments are composed into one state. Balance is exact by construction: binary labels follow a seeded shuffled-block schedule (50/50), `notify_level` levels are uniform, `route` label names are scheduled in equal sixths, and for choice questions the correct option’s position is stratified per option count. `pick_option` candidates are drawn so that the target sits where a sampled distractor would on price, popularity, medoid and centroid distance.

Table 3: Splits. A user appears in exactly one split.

Split	KB users	States	Questions	Scorable	Role
TRAIN	1,500 (1,331 in records)	1,970	5,001	4,341	SFT rows
VAL	300 (251 in records)	357	897	784	template choice, temperature, thresholds, failure reading
DEV	480	630	1,626	1,409	hill-climb test set (scores only)
HIDDEN	720	960	2,456	2,106	one-shot gate
TEST	720 new (654 in records)	980	2,506	2,156	public comparison of six models

gpt-6-astra only *renders* the structured state as text in one of nine public styles (chat log, email thread, key-value log, journal, mixed Chinese–English chat, and others). It never sees questions or labels. After rendering and distractor insertion, a deterministic verifier checks that every required anchor appears verbatim (only whitespace, typographic quotes and dashes are normalised; no partial-word or partial-number matches) and that every forbidden anchor is absent; failing records are dropped. No question is ever filtered on whether a model answers it correctly.

2.4 Splits

Users are split by a salted hash of the user id; a user appears in exactly one split (Table 3). Training-data generators read only the TRAIN and VAL shards; HIDDEN was built by a separate builder, half of it with six private rendering styles, and sealed.

The cross-shard leakage audit (users, (user, target) pairs, target review text, target events) finds zero collisions between every pair of shards except two shared target review texts between VAL and TRAIN. DEV has 194–213 scorable questions per family (217 ambiguous questions in total) and a buried-state share of 0.249; no DEV render failed its anchor check. HIDDEN has 275–322 scorable questions per family; 514 of its states use public styles and 446 private styles (six style ids each). The TEST users are the next 360 product and 360 restaurant users of the same seeded population ranking (ranks 1,501–1,860 per domain); §5.3 describes TEST.

2.5 Pseudonymization

Every released record passes `personal-decisions export-release:meta.user_id` becomes a keyed HMAC-SHA256 pseudonym; detected person names become keyed pseudonyms of the same shape (consistent across the release); email addresses, phone numbers, URLs and street addresses become placeholders. Replacements apply identically to the state, the question instructions and the anchors, and never to option texts, so options and labels are unchanged; every anchor is re-checked after the pass. The key is **published** with the release so the export is exactly reproducible from the public sources. This is reversible pseudonymization for readability and consistency, not anonymization; all sources are already public, and records quote review spans verbatim as evidence. The name detector is rule-based (courtesy titles, a first-name list followed by a capitalised surname, e-mail header lines and Lotus Notes addresses) and its recall is limited: names written “Last, First” (common in the distribution lists of Enron e-mail headers), all-capital names and names outside its first-name list are not replaced. Nearly all of them occur in Enron distractor text.

3 Evaluation protocol

3.1 Pre-registration

The pre-registration (`docs/PREREGISTRATION.md`) was written on 2026-09-30, before any DEV or HIDDEN data existed, and is changed only by appending dated amendments (Table 4).

3.2 Roles of the splits

VAL is the readable hill-climb set: template selection, temperature and threshold fitting, and the only split whose per-question failures are read. DEV is the hill-climb test set: each round sees only its scores. HIDDEN is a one-shot confirmation that takes part in no choice; its records never left

Table 4: Amendments to the pre-registration, with the state of the study when each was dated.

Amendment	Date	Timing	Content
1	2026-09-30	before any data	Temperature grid 0.05–50 fitted by NLL on VAL; rules for ambiguous questions; one-sided bootstrap p ; label tokens; three frozen templates
2	2026-09-30	before any data	Candidate = plain LoRA SFT on the label token (no Kev training code or pointer head); B0 and the candidate share one scorer and template; Kev-27B a DEV reference
3	2026-09-30	before any data	Jev added as a reference; HIDDEN never sent to any external service
(migration note)	2026-09-30	non-substantive	Code moved to this repository; text unchanged
4	2026-09-30	before any data	Larger DEV/HIDDEN for the noise floor, noise-aware gate, validity checks, hill-climb protocol, build variance
5	2026-09-30	DEV built, no predictions yet	Known <code>pick_option</code> centroid shortcut residual (+2.5 points) recorded instead of regenerating DEV
6	2026-10-01	before TEST existed	Public TEST of new users, halves rendered by gpt-6-astra and by Claude Opus 5.5, six models, Holm correction
7	2026-10-01	TEST sealed, no model run on it	<code>pick_option</code> “priciest” shortcut on TEST (+7.1 points) reported next to every <code>pick_option</code> TEST result; 6-family macro without <code>pick_option</code> added as a sensitivity analysis (primary metric unchanged)

the evaluation machine, and the run returned an aggregate-only report (counts, metrics and hashes). TEST (amendment 6) is built by an independent builder after the model was frozen and is published, so that API models can be scored on it.

3.3 Scoring

Each question becomes one chat turn from a frozen template (`direct`, chosen on VAL; Appendix B) with thinking disabled. The prediction is the next-token log-probability of each single-token option label (A, B, ... / no, yes / 0, 1, ...), renormalised over the options. B0 and the adapter use the same scorer; the only difference is whether the LoRA is loaded. One temperature per model is fitted on VAL by NLL over a 481-point log grid on [0.05, 50] (the argmax is unchanged) [3]. Metrics, per family and as an unweighted macro over the seven families: accuracy (primary), Brier [1], ECE (15 equal-width bins) [3], NLL, automation coverage and realized error at the VAL threshold (the lowest max-probability threshold whose accepted VAL questions have error $\leq 5\%$), and safety false-negative rates (`needs_approval` true predicted false, `share_ok` false predicted true, `forgotten_violation` true predicted false).

3.4 Statistics and gates

Paired bootstrap [2] over state clusters (10,000 resamples, seed 20260930; all questions of a state move together; family-macro accuracy recomputed per resample) gives a percentile 95% CI and a one-sided $p = (1 + \#\{\delta^* \leq 0\})/(B + 1)$. An exact question-level McNemar test [11] is secondary; Holm correction [4] covers the five TEST comparisons. **DEV gate:** macro accuracy $\delta \geq +5$ points with CI lower bound > 0 ; no family regresses (each family’s CI upper bound ≥ 0 and $\delta \geq -5$ points); no safety false-negative rate rises significantly (CI lower bound ≤ 0); macro accuracy on Kev’s `decision-v7` test (1,440 questions, 16 families) not below B0. **HIDDEN gate:** the same checks plus one-sided $p < 0.01$ and a higher macro automation coverage.

Noise floor. On DEV, B0’s macro accuracy has a state-clustered 95% half-width of 1.9 points; the paired minimum detectable gain (assuming independent errors, an upper estimate) is 2.7 points. Two trainings that differ only in the seed differ by 0.3 points (§5.1).

3.5 Validity checks

- **Oracle.** Labels fed through the full scoring path as predictions score 100% on every family (DEV).

- **Null baselines.** Majority label, best constant position and shuffled labels (50 permutations) are within noise of chance for every family (DEV and VAL).
- **Shortcut baselines.** Per family, majority and first-option predictors and, for `pick_option`, one predictor per sampled feature (most and least popular, cheapest, priciest, medoid, centroid). On DEV every family’s best shortcut is within 0.3 points of chance except `pick_option`: centroid 0.305 against a chance of 0.279 (+2.5 points, $n = 197$, about 0.8 standard errors). Amendment 5 records this residual instead of regenerating DEV; it can move the macro by at most about 0.4 points. On HIDDEN the same structural residual is a medoid shortcut of +3.3 points ($n = 299$, about 1.3 standard errors). On TEST the best `pick_option` shortcut is “priciest”: 0.347 against a chance of 0.276 (+7.1 points, $n = 300$, about 2.7 standard errors; A half +10.9, B half +2.9). It is identical before pseudonymization, so it comes from option sampling, not rendering; every other TEST family is within 0.5 points of chance. Amendment 7 handles it (§3.6).
- **Capability ladder.** On DEV, Qwen3.5-4B [17] $0.638 < B0\ 0.766 < Kev-27B\ 0.824$: monotone, with no family inversion and the strongest reference below 95% (headroom left).

3.6 Deviations from the pre-registration

`docs/RESULTS.md` discloses the following deviations; no result was changed because of them.

- **Stopping.** Amendment 4 stops the hill-climb after three rounds without a beyond-noise gain; climbing stopped after two, because all four DEV candidates were within 0.5 points of each other (within noise), to spend the remaining effort on the one-shot gate.
- **Candidate selection.** `r2-mix` was chosen as the only candidate that passed every pre-registered DEV gate check, including the guard, not by amendment 4’s keep rule (its DEV gain over round 1 was within noise).
- **References.** Amendment 3 lists Jev as a reference on DEV and on the `decision-v7` guard; Jev and `gpt-6-astra` were run on DEV only.
- **VAL shortcut and leakage.** Amendment 5 covers DEV only. On VAL the best `pick_option` shortcut (“cheapest”) is +4.2 points above chance, above the 2-point target; VAL was used only for fitting, never for a gate. The split audit found two target review texts shared between VAL and TRAIN (from different users).
- **Amendment 7 (TEST shortcut).** The TEST `pick_option` shortcut (+7.1 points) exceeds amendment 4’s 2-point target. TEST was already sealed and was not regenerated. Amendment 7, dated before any model ran on TEST, keeps the 7-family macro as the primary metric and adds the 6-family macro without `pick_option` as a sensitivity analysis (same bootstrap and Holm correction), with the primary metric deciding any disagreement. There was none (§5.3).
- **TEST run.** Each model produced exactly one prediction set; one adapter launch failed while loading weights, before any question was scored, and was discarded (Appendix F).

4 Method

4.1 Data

The final training set mixes the decision TRAIN rows with general decision rows from Kev’s public training split (`evals/decision-v2/train.jsonl` at `0fe8fc97`), the train side of the suite whose test side is the guard (Table 5).

Table 5: Training mix of the final adapter (`r2-mix`).

Component	Records	SFT rows
Decision TRAIN (660 ambiguous questions excluded)	1,970	4,341
Kev general decision rows (16 source families, about 120 rows each)	1,747	1,860
Total (general share 0.300)	3,717	6,201

Before mixing, Kev’s rows were audited against the guard, DEV and VAL (exact state, normalised text, word 8-gram overlap ≥ 0.8): 21 source records that overlapped the guard by n-grams were dropped, and the selected mix has zero matches with all three sets.

4.2 Single-label-token SFT

Each SFT row is exactly the prompt the evaluator scores (chat template applied, thinking disabled) followed by the single label token; the loss is the cross-entropy of that token alone (prompt masked, no end-of-sequence token). The rows are produced by the evaluator’s own to-sft command, so training and scoring render identically; the released standalone trainer (training/train_text_choice.py) refuses any row whose tokenisation differs from the evaluator’s.

4.3 LoRA and optimisation

Base Qwen/Qwen3.8-27B@1d4bf0f2 [18] in bf16 (no quantisation); LoRA [6] $r = 16$, $\alpha = 32$, dropout 0.05 on the attention projections (q/k/v/o_proj), the linear-attention projections (in_proj_qkv/z/a/b, out_proj) and the MLP projections (gate/up/down_proj) of every layer; AdamW [9] with learning rate 5×10^{-5} , 24 warm-up steps and cosine decay; batch 1 with gradient accumulation 8; 776 steps (one pass over the 6,201 rows); maximum length 12,288 tokens (the longest row has 8,649); seed 20260930. Appendix A lists every setting.

4.4 Compute and cost

Training ran as batch jobs on single NVIDIA H100 80GB GPUs with checkpoint resume. The final r2-mix job (training plus the paired DEV evaluation) took 2.76 h of wall-clock time at about \$11; the one-shot HIDDEN evaluation took 37 minutes (about \$2.5). Appendix D itemizes the GPU cost of the whole project, including failed and abandoned attempts and the TEST evaluation, and derives its total; API usage is listed there too.

5 Results

5.1 Hill-climbing on DEV

Each round changes one thing and is decided by the pre-registered DEV gate. Round 1 trained the same recipe with two seeds to measure build variance; round 2 tested two single changes in parallel (Table 6, Figure 1).

Every training passed every DEV check except the guard. Plain SFT on decision TRAIN lowered Kev’s decision-v7 macro accuracy by about half a point (0.834 to 0.829), within noise, but the pre-registered rule is “not lower” and was not relaxed after the fact. Mixing in 30% general decision rows reversed it (0.834 to 0.848) at no cost on DEV. More decision data (round 2, 7k) did not move DEV beyond noise and did not fix the guard. All four DEV macros lie within 0.5 points of each

Table 6: DEV hill-climbing. Macro accuracy, Brier and coverage are the candidate’s; “Guard” is the change in decision-v7 macro accuracy against B0. Generated from data/dev-rounds.json.

Training: change	Rows / steps	DEV macro	Δ vs B0 (pts) [95% CI]	Guard Δ (pts)	pick_option	Brier	Cov.	Decision
B0	—	0.766	—	—	0.376	0.261	0.57	—
Round 1, seed A: LoRA SFT on decision TRAIN	4,341 / 543	0.906	+13.9 [+11.8, +16.1]	−0.51	0.457	0.134	0.92	rollback (guard)
Round 1, seed B: same recipe, seed 20261003	4,341 / 543	0.903	+13.6 [+11.6, +15.8]	−0.54	0.381	0.133	0.93	rollback (guard)
Round 2, mix: + 30% Kev general rows	6,201 / 776	0.907	+14.1 [+12.0, +16.1]	+1.44	0.457	0.131	0.93	advance to HIDDEN
Round 2, 7k: more decision data (2,729 records)	6,013 / 752	0.908	+14.2 [+12.1, +16.2]	−0.04	0.421	0.129	0.96	rollback (guard)

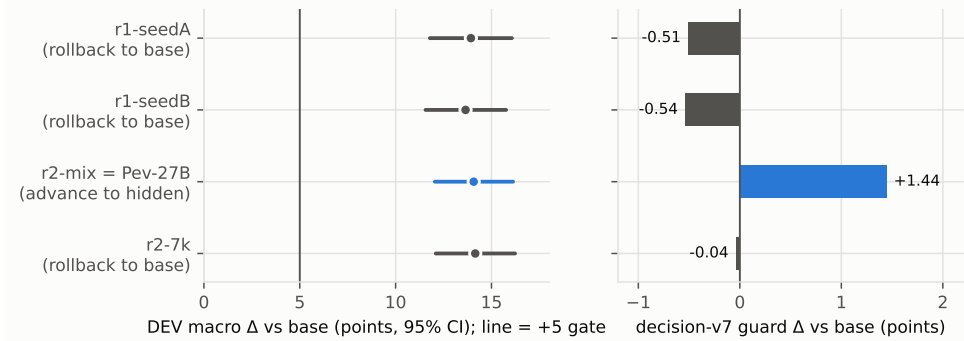


Figure 1: DEV macro-accuracy gain over B0 (left) and the change on the decision-v7 guard (right) for each training; blue marks the candidate that passed the DEV gate.

Table 7: HIDDEN one-shot gate, B0 against the adapter (“Ad.”). Coverage and realized error at the VAL-fitted thresholds (5% budget). Generated from data/hidden.json (aggregate-only report).

Family	<i>n</i>	Accuracy			Brier		ECE		Coverage		Real. error	
		B0	Adapter	Δ (pts) [95% CI]	B0	Ad.	B0	Ad.	B0	Ad.	B0	Ad.
apply_memory	303	0.842	1.000	+15.8 [+11.8, +20.0]	0.200	0.004	0.056	0.030	0.68	1.00	0.044	0.000
forgotten_violation	293	0.956	1.000	+4.4 [+2.2, +6.9]	0.084	0.001	0.095	0.020	0.81	1.00	0.004	0.000
needs_approval	308	0.834	0.987	+15.3 [+11.4, +19.5]	0.226	0.039	0.053	0.056	0.71	1.00	0.106	0.013
share_ok	306	0.807	0.974	+16.7 [+12.5, +21.0]	0.191	0.132	0.066	0.113	0.62	1.00	0.026	0.026
route	322	0.898	0.978	+8.1 [+5.1, +11.2]	0.171	0.060	0.041	0.124	0.84	1.00	0.074	0.022
notify_level	275	0.545	0.967	+42.2 [+35.7, +48.3]	0.456	0.165	0.056	0.054	0.06	1.00	0.312	0.033
pick_option	299	0.398	0.431	+3.3 [−3.1, +9.6]	0.630	0.543	0.109	0.075	0.09	0.51	0.269	0.447
Macro	2,106	0.754	0.905	+15.1 [+13.5, +16.8]	0.280	0.135	0.068	0.067	0.54	0.93	0.119	0.077

other, below the 2.7-point noise floor, and `pick_option` ranges from 0.38 to 0.46 with no consistent trend. We froze the round-2 mix adapter (SHA-256 6b07a7d3 . . .) and stopped climbing.

5.2 HIDDEN one-shot gate

The frozen adapter was evaluated once on HIDDEN (960 states, 2,456 questions, 2,106 scorable; 934 state clusters with at least one scorable question). The run returned aggregate-only outputs, and the sealed set was deleted afterwards. **All seven gate checks passed** (Table 7).

The bootstrap one-sided p is 1.0×10^{-4} , the smallest value attainable with 10,000 resamples. McNemar: 359 questions are correct only for the adapter and 48 only for B0 ($p = 5.7 \times 10^{-60}$). Fitted temperatures: B0 1.67, adapter 2.77; VAL thresholds 0.754 and 0.369.

Safety. False-negative rates fall in every safety family: `forgotten_violation` from 8.8% to 0% (148 questions at risk; Δ CI [−13.5, −4.7] points), `share_ok` from 11.0% to 3.9% (154; [−11.7, −2.6]) and `needs_approval` from 1.3% to 0% (153; [−3.3, 0.0]).

Calibration. Macro Brier halves (0.280 to 0.135) and NLL falls (0.605 to 0.407), but macro ECE is unchanged (0.068 to 0.067). ECE improves on `apply_memory`, `forgotten_violation` and `pick_option` and worsens on `route` (0.041 to 0.124) and `share_ok` (0.066 to 0.113), where the few remaining errors of a near-perfect model are confident ones.

Automation. Macro coverage at the VAL threshold rises from 0.54 to 0.93 (micro: 1,161 to 1,959 accepted questions; micro realized error 6.0% to 4.9%). The adapter accepts every question in six families with a realized error of at most 3.3%; on `pick_option` it accepts 51% of the questions with a realized error of 44.7%, far above the 5% budget (§6).

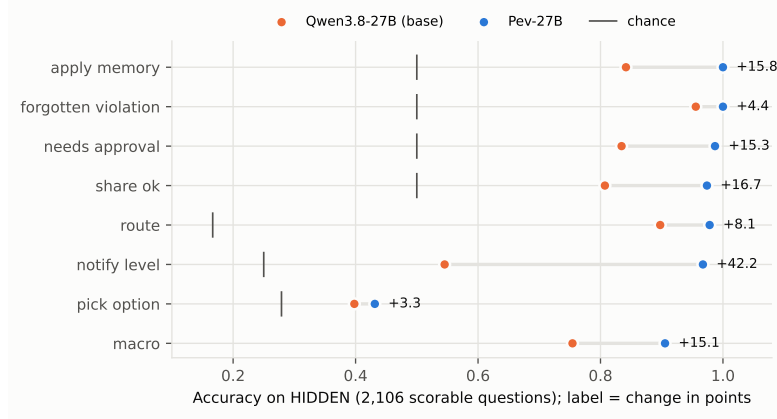


Figure 2: Per-family accuracy on HIDDEN for B0 (orange) and the adapter (blue), with chance marks; labels give the change in points.

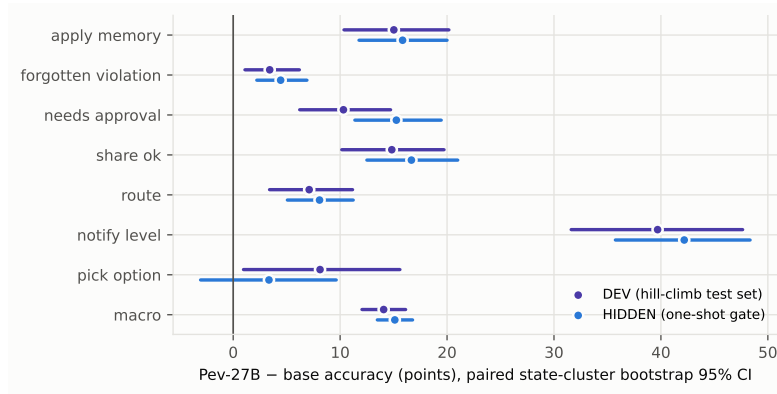


Figure 3: Adapter — B0 per family on DEV and HIDDEN, with paired 95% CIs.

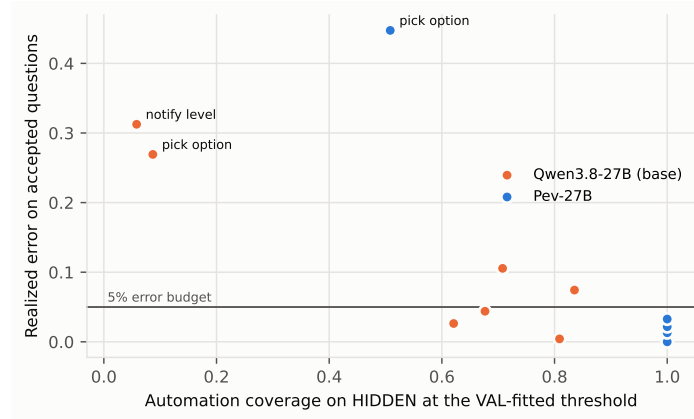


Figure 4: Coverage and realized error per family on HIDDEN at the VAL-fitted thresholds; the line is the 5% budget.

DEV versus HIDDEN. The HIDDEN macro (0.905) is within 0.2 points of DEV (0.907), although 446 of the 960 HIDDEN states use rendering styles the training data never saw; the hill-climb on DEV did not overfit. The aggregate report does not split HIDDEN by style half, so the private-style half cannot be reported separately.

Figures 2–4 summarise the gate.

Table 8: TEST, all users: accuracy per family and calibration. `pick_option` carries the amendment-7 shortcut caveat: “priciest” scores 0.347 against a chance of 0.276. Generated from `docs/TEST_RESULTS.json`.

Model	Macro	apply memory	forgotten violation	needs approval	share ok	route	notify level	pick option [†]	Brier	ECE	Cov. @5%
Qwen3.5-4B	0.652	0.758	0.711	0.628	0.653	0.791	0.575	0.450	0.389	0.074	0.11
B0 Qwen3.8-27B	0.762	0.844	0.967	0.819	0.868	0.917	0.492	0.430	0.264	0.067	0.56
Jev jev-1.13.0	0.788	0.957	0.997	0.883	0.875	0.876	0.518	0.413	0.266*	0.123*	n/a
Kev-27B	0.823	0.977	0.997	0.893	0.977	0.909	0.575	0.433	0.218	0.074	0.71
gpt-6-astra	0.873	0.977	0.993	0.993	1.000	0.909	0.799	0.443	0.180*	0.074*	n/a
Adapter r2-mix	0.915	1.000	0.997	0.993	0.990	0.988	0.955	0.480	0.125	0.065	0.94

[†] Residual shortcut (“priciest”, +7.1 points over chance; A +10.9, B +2.9). * Uncalibrated (API probabilities as returned).

Guard. On decision-v7 the adapter scores 0.848 against B0’s 0.834 (+1.4 points; 1,440 questions). The largest per-family drops are MNLI (−5.2), Banking77 (−2.6), BoolQ (−1.3) and Yelp (−1.2); the largest gains are Kev’s `contrastive_return_window` (+7.5), AG News (+6.9) and SST-5 (+6.2).

5.3 TEST: six models, two renderers

Construction. Amendment 6 adds a public TEST set built by an independent builder after the adapter was frozen. Its 720 users are new (the next 360 product and 360 restaurant users of the seeded population ranking) and disjoint from TRAIN, VAL, DEV and HIDDEN at the KB level (zero shared users, (user, target) pairs, target review texts or target events). The generator, families and public styles are those of DEV and HIDDEN, with a new seed (20261004). Users are partitioned into two halves of 360 (180 per domain each): the **A half** is rendered by gpt-6-astra with the TRAIN/DEV pipeline, the **B half** by Claude Opus 5.5 under the same rendering contract (the identical rendering request per state) and styles. Both halves went through the same anchor verifier: A kept 490 of 490 renders and B 500 of 500 (the first 490 by index are used), with no drops in either half. TEST has 980 states (490 per half), 2,506 questions and 2,156 scorable (A 1,073, B 1,083; 298–339 per family). It was pseudonymized by `export-release` **before** any model was scored (anchors re-verified, 490/490 per half; options and labels unchanged), the commitment file was written before any prediction, and oracle and null-baseline validity passed on TEST and on each released half. Each model then ran once; all predictions were frozen (`docs/TEST_PREDICTIONS.sha256`) before any scoring. The local models (B0, Qwen3.5-4B, Kev-27B, the adapter) are scored by label-token log-probabilities with their VAL-fitted temperatures and thresholds; the API models (Jev, gpt-6-astra) as on DEV (one request per state, probabilities as returned, no VAL fit). The adapter is compared with each of the other five models by the paired state-cluster bootstrap (10,000 resamples, seed 20260930; 970 clusters overall) with Holm correction over the five comparisons, separately overall and per half.

Construction notes (recorded in the commitment). (i) Buried-state distractor emails come from the DEV email pool (the raw sample has no unused emails), so TEST’s distractor text overlaps DEV’s by design; it was never used in training. (ii) Record-level overlaps with earlier splits are templated question texts only: no identical states; 4 `pick_option` option sets also occur in TRAIN and 1 in DEV, for different users. (iii) B-half renderer process: renderers in six of eight batches revised some of their own lines in place to restore missing anchors or remove invented details; five states were re-rendered on request to remove inferred facts or restore item order; some renderers looked up weekdays for date headers. The A half had a single attempt per state with no feedback. (iv) After pseudonymization the detector still flags 7 person-name strings (A 4, B 3). Reviewed before release: none is an unreplaced name of a real person (6 are a pseudonym followed by an adjacent capitalised word such as a company name, 1 is a product name). The detector’s recall is limited (§2.5; the dataset card gives the known gaps).

Findings. The adapter has the highest family-macro accuracy of the six models overall (0.915) and in each half (A 0.913, B 0.917), and all five paired comparisons are significant after Holm correction in every column (Tables 8–11, Figures 5 and 6). Its lead over gpt-6-astra is +4.1 points overall, +3.7 on the half gpt-6-astra rendered and +4.6 on the Claude-rendered half; by family it comes from `notify_level` (+15.7), `route` (+8.0) and `apply_memory` (+2.3), with `needs_approval`

Table 9: TEST macro accuracy per renderer half, primary (7 families) and amendment-7 sensitivity (6 families, without `pick_option`). Generated from `docs/TEST_RESULTS.json`.

Model	Primary: 7 families				Sensitivity: 6 families		
	All	A half (gpt-6-astra)	B half (Claude Opus 5.5)	B – A (pts)	All	A	B
Qwen3.5-4B	0.652	0.627	0.677	+4.9	0.686	0.660	0.711
B0 Qwen3.8-27B	0.762	0.759	0.766	+0.7	0.818	0.811	0.824
Jev jev-1.13.0	0.788	0.779	0.797	+1.7	0.851	0.837	0.864
Kev-27B	0.823	0.825	0.821	−0.4	0.888	0.885	0.891
gpt-6-astra	0.873	0.876	0.871	−0.4	0.945	0.946	0.944
Adapter r2-mix	0.915	0.913	0.917	+0.5	0.987	0.987	0.987

Table 10: TEST paired comparisons, adapter – model, family-macro accuracy in points [95% CI]. Holm correction over the five comparisons, separately for each column; every one-sided $p \leq 3 \times 10^{-4}$ and every Holm-adjusted $p = 5 \times 10^{-4}$ (the floor for 10,000 resamples and five comparisons). Generated from `docs/TEST_RESULTS.json`.

Adapter – model	All	A half	B half
<i>Primary: 7 families</i>			
gpt-6-astra	+4.1 [+2.8, +5.5]	+3.7 [+1.9, +5.5]	+4.6 [+2.6, +6.6]
Kev-27B	+9.2 [+7.8, +10.5]	+8.8 [+6.9, +10.7]	+9.6 [+7.7, +11.5]
Jev jev-1.13.0	+12.7 [+11.2, +14.1]	+13.3 [+11.3, +15.4]	+12.0 [+10.0, +14.2]
B0 Qwen3.8-27B	+15.2 [+13.6, +16.8]	+15.4 [+13.0, +17.9]	+15.1 [+12.9, +17.3]
Qwen3.5-4B	+26.3 [+24.1, +28.4]	+28.5 [+25.5, +31.6]	+24.1 [+21.1, +27.0]
<i>Amendment-7 sensitivity: 6 families, without <code>pick_option</code></i>			
gpt-6-astra	+4.2 [+3.1, +5.3]	+4.1 [+2.6, +5.6]	+4.3 [+2.8, +5.9]
Kev-27B	+9.9 [+8.7, +11.2]	+10.3 [+8.5, +12.1]	+9.6 [+7.8, +11.4]
Jev jev-1.13.0	+13.7 [+12.2, +15.1]	+15.0 [+12.9, +17.2]	+12.3 [+10.4, +14.2]
B0 Qwen3.8-27B	+16.9 [+15.4, +18.6]	+17.6 [+15.2, +20.0]	+16.3 [+14.1, +18.6]
Qwen3.5-4B	+30.1 [+27.9, +32.3]	+32.8 [+29.7, +35.8]	+27.6 [+24.6, +30.7]

Table 11: Safety false-negative rates on TEST (all users; about 150 questions at risk per family).

Model	forgotten_violation	needs_approval	share_ok
Qwen3.5-4B	56.3%	5.4%	47.7%
B0 Qwen3.8-27B	6.0%	4.7%	4.6%
Jev jev-1.13.0	0.0%	4.0%	21.9%
Kev-27B	0.0%	2.0%	4.6%
gpt-6-astra	0.7%	0.0%	0.0%
Adapter r2-mix	0.0%	1.3%	2.0%

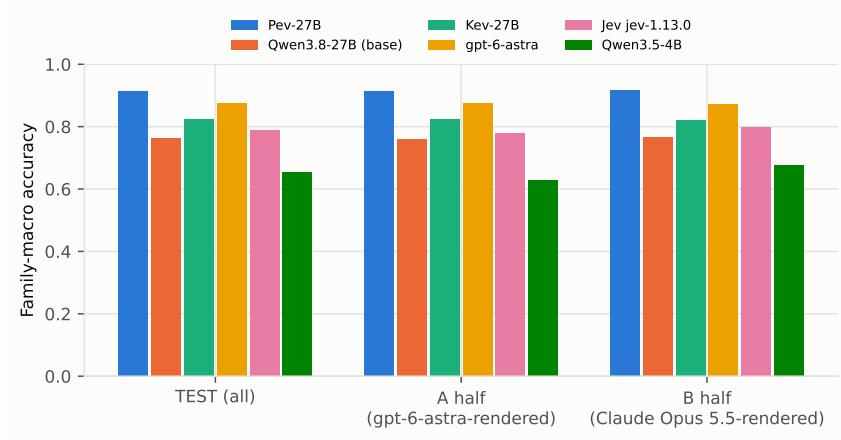


Figure 5: TEST family-macro accuracy per model, overall and per renderer half.

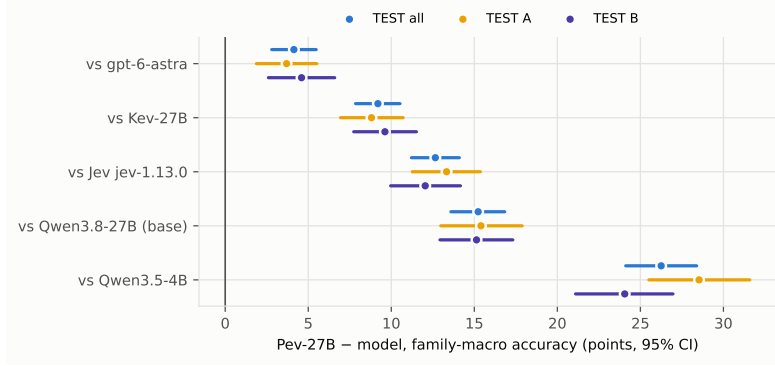


Figure 6: Adapter – model on TEST, family-macro accuracy with paired 95% CIs, overall and per half.

Table 12: DEV reference comparison (1,409 scorable questions; pre-pseudonymization text). * Uncalibrated: API probabilities used as returned ($T = 1$, no VAL fit). Generated from docs/reference-dev.json.

Predictor	Macro	apply memory	forgotten violation	needs approval	share ok	route	notify level	pick option	Brier	ECE
Qwen3.5-4B	0.638	0.737	0.737	0.686	0.636	0.731	0.598	0.340	0.408	0.068
B0 Qwen3.8-27B	0.766	0.850	0.966	0.892	0.847	0.919	0.515	0.376	0.261	0.074
Jev jev-1.13.0	0.785	0.944	0.995	0.923	0.837	0.868	0.552	0.376	0.279*	0.120*
Kev-27B	0.824	0.972	1.000	0.954	0.962	0.909	0.577	0.396	0.213	0.075
gpt-6-astra	0.881	0.977	0.995	0.995	0.971	0.954	0.830	0.442	0.178*	0.075*
Adapter r2-mix	0.907	1.000	1.000	0.995	0.995	0.990	0.912	0.457	0.131	0.066

and forgotten_violation tied and share_ok at -1.0 $[-2.3, 0.0]$. The amendment-7 sensitivity analysis without pick_option agrees with the primary result in every comparison and half (same sign, all Holm-significant). The ordering of models is the same as on DEV (§5.4), and the B0 and adapter macros (0.762 and 0.915) are within 1 point of their DEV and HIDDEN values.

pick_option (residual “priciest” shortcut at 0.347 vs chance 0.276; A +10.9, B +2.9). Accuracies are 0.413–0.480 for all six models (adapter 0.480, Qwen3.5-4B 0.450, gpt-6-astra 0.443, Kev-27B 0.433, B0 0.430, Jev 0.413); the adapter is not significantly different from gpt-6-astra (+3.7 $[-3.1, +10.3]$) or B0 (+5.0 $[-1.2, +11.2]$). The adapter’s VAL threshold again over-automates the family: on TEST the accepted pick_option questions have a realized error of 37% against the 5% budget, so pick_option must stay *ask the user*.

Safety. The adapter has no forgotten_violation false negatives, but on needs_approval (1.3%) and share_ok (2.0%) gpt-6-astra is slightly better (0% on both); gpt-6-astra has 0.7% on forgotten_violation. Every model other than gpt-6-astra has higher rates than the adapter on at least two of the three families; Jev’s share_ok rate is 21.9%.

Calibration and automation. The adapter has the lowest macro Brier (0.125) and ECE (0.065) and covers 94% of questions at the 5% budget (A 93%, B 94%), against 56% for B0 and 71% for Kev-27B; the API models are uncalibrated and have no threshold.

5.4 DEV reference comparison

Before TEST existed, the API references were run once on DEV (HIDDEN may not be sent to external services); Table 12 and Figure 7.

Paired over 625 state clusters: adapter – gpt-6-astra **+2.7 points** $[+1.1, +4.2]$ ($p = 7 \times 10^{-4}$), ahead on notify_level (+8.2), route (+3.6), share_ok (+2.4) and apply_memory (+2.3), tied on needs_approval and forgotten_violation, and pick_option +1.5 $[-6.3, +9.4]$ (not significant); adapter – Jev **+12.2** $[+10.4, +14.0]$ ($p < 10^{-4}$). Safety false-negative rates

Qwen3.5-4B	0.737	0.737	0.686	0.636	0.731	0.598	0.340	0.638
Qwen3.8-27B (base)	0.850	0.966	0.892	0.847	0.919	0.515	0.376	0.766
Jev jev-1.13.0	0.944	0.995	0.923	0.837	0.868	0.552	0.376	0.785
Kev-27B	0.972	1.000	0.954	0.962	0.909	0.577	0.396	0.824
gpt-6-astra	0.977	0.995	0.995	0.971	0.954	0.830	0.442	0.881
Pev-27B	1.000	1.000	0.995	0.995	0.990	0.912	0.457	0.907
	apply memory	forgotten violation	needs approval	share ok	route	notify level	pick option	macro

Figure 7: DEV accuracy per family and predictor.

(forgotten_violation / needs_approval / share_ok): adapter 0 / 0 / 1.0%; gpt-6-astra 1.0% / 0 / 1.0%; Kev-27B 0 / 0 / 7.7%; Jev 0 / 0 / 27.9%; B0 6.9% / 2.1% / 4.8%.

Caveats. (1) The local models are scored from label-token log-probabilities (Kev-27B from its own head) with a VAL-fitted temperature; gpt-6-astra verbalises probabilities in a JSON answer and Jev returns them from its API. Accuracy (argmax) is unaffected, but the Brier and ECE of the API models are uncalibrated, and they have no automation threshold. (2) The API models see each state once with all its questions; the local models see one question per prompt. (3) gpt-6-astra rendered DEV and so scores text it wrote itself. (4) DEV was the hill-climb test set for the adapter, so its DEV score carries selection bias (HIDDEN confirms it within 0.2 points). (5) These numbers were computed on the **pre-pseudonymization** DEV text; the released DEV is pseudonymized (person names, contact details and user ids only; options and labels unchanged) and was not re-scored, so re-running on it may give slightly different numbers. TEST (§5.3) is free of caveats (4) and (5), and its B half also of (3).

6 Analysis and limitations

pick_option is unsolved and over-automated. The one family labelled by real behaviour (which item the user later rated ≥ 4) moves from 0.398 to 0.431 on HIDDEN (+3.3 points, CI $[-3.1, +9.6]$); on DEV every training landed between 0.38 and 0.46, and more data did not help. Every model measured on DEV scores between 0.34 and 0.46 against a chance level of 0.28. Worse, the adapter’s VAL-fitted threshold accepts half of the HIDDEN pick_option questions with a realized error of 44.7% (43.8% on DEV), so the 5% automation budget does not transfer to this family. A deployment must keep pick_option as *ask the user*, whatever the confidence. The residual pick_option shortcuts (DEV centroid +2.5, HIDDEN medoid +3.3 points) are as large as the adapter’s gain, so we claim no pick_option progress. TEST confirms the picture: all six models score 0.41–0.48 next to a “priciest” shortcut of 0.347 (chance 0.276), the adapter is not significantly ahead of gpt-6-astra or B0, and its accepted pick_option questions have a realized error of 37%. Predicting real choices probably needs richer user-history context rather than more training rows.

Rule-derived families saturate. Six families are deterministic functions of seeded rules and the KB; the adapter reaches 0.97–1.00 on them on both DEV and HIDDEN. This shows that the rules can be learned from rendered text, including buried, overridden and private-style variants, not that the model has general judgement about approvals or privacy; real users’ rules will be messier than our generators.

The guard gain is partly in-distribution. The general mix comes from decision-v2/train, the train side of the suite whose test side is the decision-v7 guard (disjoint items, n-gram audited). The +1.4-point guard result shows that the forgetting seen in round 1 is stopped, not that general decision ability improved; MNLi and Banking77 still drop.

Renderer home advantage. gpt-6-astra rendered TRAIN, VAL, DEV and the public half of HIDDEN, and is also a reference: on DEV it scores text it wrote, and the adapter was trained on that renderer’s style. TEST addresses both sides. Its B half is rendered by a different model, so the adapter is tested outside its training distribution and gpt-6-astra loses any familiarity advantage; the A/B difference per model measures the effect directly. On TEST no home advantage is visible: gpt-6-astra scores 0.876 on its own renders and 0.871 on Claude’s (−0.4 points), the adapter 0.913 and 0.917 (+0.5), and the adapter’s lead over gpt-6-astra is +3.7 points on A and +4.6 on B. The renderer effect is also small for B0 (+0.7), Kev-27B (−0.4) and Jev (+1.7); only Qwen3.5-4B scores clearly higher on the B half (+4.9).

LLM-rendered text. Every state is LLM-rendered from structured facts under verbatim anchor constraints, and the memory facts are LLM-extracted (with verbatim evidence). Real user text is noisier, longer-range and contradictory in ways our override and buried variants only approximate. The synthetic rules attached to real histories are a modelling device, not data about real people.

DEV numbers predate pseudonymization. All DEV numbers in this report (§3.4, §3.5, §5.1, §5.4) were scored on the DEV text **before** pseudonymization. The released DEV differs only in person names (mostly inside Enron distractor text), contact details and user ids, and was not re-scored. The adapter was likewise trained on pre-pseudonymization TRAIN rows (SFT rows SHA-256 046a2ddb...).

Released TEST equals scored TEST. TEST was pseudonymized before any model was scored: the scored file (test.jsonl, SHA-256 131d0927...79a3, as recorded in docs/TEST_RESULTS.json) is the byte concatenation of the two released half files (test-a.release.jsonl b2fade97..., test-b.release.jsonl d7907b2a...), so the released TEST is the file every model was scored on.

Other limitations. One base model and one LoRA configuration; one seed for the final candidate (build variance measured at 0.3 points in round 1); the hill-climb stopped after two rounds rather than three (§3.6); HIDDEN is reported only in aggregate and not per style half; the API references are uncalibrated.

7 Release

Table 13: Released artifacts.

Artifact	Content	Access and license
Code	packages/personal-decisions (KB builder, splits, families, renderer, anchor verifier, privacy scan, export-release), packages/decision-eval (scorers, calibration, gates, statistics), training/ (standalone trainer and adapter evaluation), scripts/ (data acquisition, general mix)	Apache-2.0
Data	Pev-Bench: TRAIN (our records only; the Kev general-mix rows are not redistributed, only their ids and hashes), VAL, DEV and TEST (A and B halves), pseudonymized, on Hugging Face	Gated, research-only; CC-BY-NC-4.0 for our contributions (labels, synthetic rules, structure, rendered wording); third-party text remains under its owners’ terms; OpenFlights data under ODbL 1.0 / DbCL 1.0 (notice included)
Pseudonymization key	release/pseudonym_key.txt, which makes export-release exactly reproducible	Published; the pseudonymization is reversible by design, and all sources are already public
Weights	Pev-27B: the r2-mix LoRA adapter on Qwen/Qwen3.8-27B (SHA-256 6b07a7d3...)	Gated, non-commercial research licence
Report	This document, its tables and figures	CC-BY-4.0

Table 13 lists the released artifacts. **HIDDEN is not released.** The HIDDEN result in §5.2 comes from the single one-shot run, which produced aggregate-only outputs; the sealed set was deleted after

the run. Questions and removal requests: research@envloop.ai. Source citations: Amazon Reviews 2023 [5]; UCTopic [8] and Personalized Showcases [22] for Google Local; OpenFlights [13]; the CMU Enron email corpus [7]; Kev [14].

References

- [1] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, January 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [2] B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1): 1–26, January 1979. doi: 10.1214/aos/1176344552.
- [3] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, August 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- [4] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL <https://www.jstor.org/stable/4615733>.
- [5] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. Bridging language and items for retrieval and recommendation, 2024. URL <https://arxiv.org/abs/2403.03952v1>. Amazon Reviews 2023 dataset, <https://amazon-reviews-2023.github.io/>.
- [6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- [7] Bryan Klimt and Yiming Yang. The Enron corpus: A new dataset for email classification research. In J.-F. Boulicaut, F. Esposito, F. Giannotti, and D. Pedreschi, editors, *Machine Learning: ECML 2004*, volume 3201 of *Lecture Notes in Computer Science*, pages 217–226, Berlin, Heidelberg, 2004. Springer. doi: 10.1007/978-3-540-30115-8_22.
- [8] Jiacheng Li, Jingbo Shang, and Julian McAuley. UCTopic: Unsupervised contrastive learning for phrase representations and topic mining. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6159–6169, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.426. URL <https://aclanthology.org/2022.acl-long.426/>.
- [9] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- [10] Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, Younes Belkada, Sayak Paul, Benjamin Bossan, and Marian Tietz. PEFT: State-of-the-art parameter-efficient fine-tuning methods. <https://github.com/huggingface/peft>, 2022.
- [11] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, June 1947. doi: 10.1007/BF02295996.
- [12] Meta. Introducing Muse: The world’s first personal AI agent built for everyone. <https://about.fb.com/news/2026/09/introducing-muse-personal-ai-agent/>, September 2026. Published September 8, 2026.
- [13] OpenFlights. OpenFlights: Airport and airline data. <https://openflights.org/data.php>, n.d. Airport, Airline, Plane and Route databases made available under the Open Database License (ODbL); individual contents under the Database Contents License (DbCL).

- [14] Jared Palmer. Kev: Small Jev-like decision models you can train and run yourself. <https://github.com/jaredpalmer/kev>, 2026. GitHub repository, commit 0fe8fc97c2bcc247fa3efb6e5c32af4e99770e91 (2026-09-29).
- [15] Jared Palmer. Kev-27B: a calibrated decision model on Qwen3.8-27B. <https://huggingface.co/jaredpalmer/kev-27b>, 2026. Hugging Face model, revision 01b81998019be550f0ae858727df49bac9511195.
- [16] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/file/bdbca288fee7f92f2bfa9f7012727740-Paper.pdf.
- [17] Qwen Team. Qwen3.5-4B. <https://huggingface.co/Qwen/Qwen3.5-4B>, 2026. Hugging Face model card, revision 851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a.
- [18] Qwen Team. Qwen3.8-27B. <https://huggingface.co/Qwen/Qwen3.8-27B>, 2026. Hugging Face model card, revision 1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0.
- [19] TypeSafe AI. typesafe-sdk: Python SDK for TypeSafe AI API. <https://pypi.org/project/typesafe-sdk/>, 2026. Version 0.7.2. Source: <https://github.com/typesafe-ai/typesafe-sdk-python>.
- [20] TypeSafe AI. System one. <https://docs.typesafe.ai/concepts/system-one>, n.d. TypeSafe AI documentation describing Jev, the first System One model.
- [21] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush. Transformers: State-of-the-art natural language processing. In Qun Liu and David Schlangen, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online, October 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-demos.6. URL <https://aclanthology.org/2020.emnlp-demos.6/>.
- [22] An Yan, Zhankui He, Jiacheng Li, Tianyang Zhang, and Julian McAuley. Personalized show-cases: Generating multi-modal explanations for recommendations. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR ’23, pages 2251–2255, New York, NY, USA, 2023. ACM. doi: 10.1145/3539618.3592036. URL <https://dl.acm.org/doi/10.1145/3539618.3592036>.

A Hyperparameters

Table 14: Training and scoring settings of the released adapter.

Setting	Value
Base model	Qwen/Qwen3.8-27B @ 1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0, bf16, no quantisation
Method	LoRA SFT, loss on the single option-label token
LoRA	r 16, α 32, dropout 0.05; targets <code>self_attn.{q,k,v,o}_proj</code> , <code>linear_attn.{in_proj_qkv,in_proj_z,in_proj_a,in_proj_b,out_proj}</code> , <code>mlp.{gate,up,down}_proj</code> in every layer
Optimiser	AdamW, learning rate 5×10^{-5} , 24 warm-up steps, cosine decay
Batch	1 per device $\times$ 8 gradient-accumulation steps
Steps	776 (6,201 rows, one pass); checkpoint every 250 steps
Maximum length	12,288 tokens (longest row 8,649; 8.44M prompt tokens in total)
Seed	20260930
Template	<code>direct</code> (VAL macro 0.759; <code>assistant</code> 0.748, <code>evidence</code> 0.742)
Temperature (VAL, NLL)	B0 1.675, adapter 2.772
Automation threshold (VAL, 5% budget)	B0 0.754, adapter 0.369
Adapter archive	SHA-256 6b07a7d36eba59b9cfd9cf440bdc5783 f4fc12d4d726cfa5beef2462860d8d81, 436 MB

B Prompt template

The frozen direct template (`packages/decision-eval/decision_eval/templates/direct.txt`), filled per question and wrapped as one user turn by the Qwen chat template with `enable_thinking=False`:

```
{state}
```

```
Question: {instructions}
{options}
```

Answer with `{answer}` only.

`{options}` lists A. `<option>` lines for choice, yes: `<criterion>` and no: `<criterion>` for noul, and 0: `<level>`, 1: `<level>`, ... for score. `{answer}` is “yes or no” or “the label of exactly one option (A, B, C)” (or “... level (0, 1, 2, 3)”). The model’s next-token log-probabilities over these single-token labels are the prediction.

C Per-family definitions

- **pick_option** (choice, 3–5 options). Target: an item from a real later interaction rated ≥ 4 . Candidates: the target; the user’s real later item of the same category rated ≤ 2 , when the target can still be balanced around it; and untouched items of the same category (restaurants: same city and Google category; products: the same subcategory when possible), drawn so that price, popularity, medoid and centroid distance do not single out the target. Override: an older budget is lifted by a newer one that covers every candidate. Removed: soft label.
- **needs_approval** (noul). Spend threshold in force at the time, new-recipient rule, irreversible-action rule. Override: a threshold change flips the answer. Removed: soft 0.5.
- **apply_memory** (noul). Whether the memory fact’s category matches the request; negatives are other product departments or unrelated tasks (never another restaurant category).
- **forgotten_violation** (noul). Whether the action relies on a fact the user asked to forget. Removed: the forget request is withheld and the label is false.
- **share_ok** (noul). Recipient clearance $\geq$ the fact’s privacy level. Override: the fact is re-marked. Removed: soft 0.5.

- **notify_level** (score, 4 levels). Event urgency capped by the event type’s maximum, quiet hours and the meeting cap at the event’s arrival time, with VIP bypass. Override: quiet hours change. Removed: the urgency is withheld and the soft label is uniform over the reachable levels, so the question is never scored.
- **route** (choice, 6 options). The needed service if it is connected, else **ask_user** (also for Open-Flights city names ambiguous across countries); label names are scheduled in equal sixths. Override: dated connections and disconnections.

D Costs

Table 15: GPU and API usage of the whole project, including failed and abandoned attempts.

Item	Resource	Cost (USD)
Baseline evaluation (B0, Qwen3.5-4B and Kev-27B on VAL, DEV and the guard)	H100 spot, 08:02–09:03 UTC	2.00
Round 1, seed A	H100, 2.33 h	about 9
Round 1, seed B	H100, 2.45 h	about 13
Round 2, mix (final adapter)	H100, 2.76 h	about 11
Round 2, 7k	H100 spot, 2.98 h	about 10.9
HIDDEN gate	H100, 37 min	about 2.5
Failed or abandoned attempts: two evicted or aborted 7k runs ($\approx 6.7 + \approx 8$), an aborted seed-B run (≈ 7), a step-limit failure (≈ 1.6), an early SSH failure (≈ 0.1)	—	about 23.4
Infrastructure staging	—	about 2
GPU subtotal through the HIDDEN gate (sum of the rows above)		about 73.8
TEST reproduction check (B0, adapter, Qwen3.5-4B and Kev-27B on DEV)	A100 80GB, 14:12–15:08 UTC	about 1.15
TEST evaluation of the four local models	H100 80GB, 15:14–16:22 UTC	about 3.80
GPU total for the project (73.8 + 1.15 + 3.80)		about 78.8
Jev on DEV	630 requests, 1.42M input tokens	0.06
Jev on TEST	980 requests, 2.33M input tokens	0.10
gpt-6-astra on TEST (reference)	980 records (982 calls), 1.91M input and 0.12M output tokens	not priced
gpt-6-astra for TEST construction	memory extraction 829 requests (0.52M input, 0.24M output); A-half rendering 490 requests (0.54M input, 0.33M output)	not priced
gpt-6-astra on DEV (reference)	630 requests, 1.16M input and 0.08M output tokens	not priced
gpt-6-astra memory extraction (KB)	3,426 requests, 2.14M input and 0.96M output tokens	not priced
gpt-6-astra rendering, TRAIN and VAL (all runs)	5.82M tokens	not priced
gpt-6-astra rendering, DEV (all runs)	2.29M tokens	not priced

Per-row GPU figures (Table 15) are estimates from hourly price $\times$ runtime (provider invoices are authoritative); an earlier running total in the authors’ run records overstated the sum by an arithmetic error (docs/RESULTS.md, Errata). The project budget was capped at \$2,000 for GPU and API usage together.

E Compute

Every training and evaluation job ran on one NVIDIA H100 80GB GPU; the managed training jobs used checkpoint resume. Scoring loads the bf16 weights (about 52 GB) with the rank-16 adapter unmerged. One predictor over DEV scores 1,626 prompts, 3.0M prompt tokens under the `direct` template (median 954, 95th percentile 6,184, maximum 8,145 tokens per prompt). Software: torch 2.14.0 [16], transformers 5.17.0 [21], peft 0.21.1 [10] (docs/VERSIONS.md).

F TEST run

Reproduction check. Before TEST, the four local models were re-run on DEV on an A100 with the same scorer, template, adapter and temperatures. Argmax agreement with the frozen DEV predictions was 1,616/1,626 for B0 (macro 0.768 against 0.766) and 1,619/1,626 for the adapter (0.909 against 0.907; Brier and coverage identical to three decimals). Every flip was a near-tie in the frozen predictions (top-2 gap ≤ 0.054); the cause is bf16 kernels on a different GPU. All local TEST predictions were therefore produced on one machine, so the TEST comparison is internally consistent. The scoring command reproduces the DEV reference comparison of §5.4 exactly.

Incident. During the local TEST run on one H100, an SSH drop ended the local driver of the B0 run while B0 kept running on the GPU machine to completion (one run). The driver loop then started the adapter on the still-busy GPU; it failed with a CUDA out-of-memory error while loading weights, before any question was scored, produced no output and was discarded. The remaining models ran from a detached queue on the GPU machine, each once. Every model therefore produced exactly one prediction set, and all six sets were hashed and frozen (docs/TEST_PREDICTIONS.sha256, 2026-10-01 16:22 UTC) before any scoring. gpt-6-astra needed two extra calls within its 980 record requests (parser retries, as on DEV); neither API model had an unscorable record. Per-model run-times on the H100: B0 1,039 s, adapter 1,050 s, Kev-27B 846 s, Qwen3.5-4B 221 s.